

ELD Trip Planner

Project overview · Full Stack Developer assessment, Spotter AI
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Live app	add the hosted URL
Source code	add the GitHub URL
Walkthrough video	add the Loom URL

What it does

The app takes a truck trip and returns a plan the driver can legally drive, with the daily logs already filled out.

Inputs	Outputs
Current location, pickup location, dropoff location, current cycle used (hrs)	Route map with every stop and rest, trip schedule, and one drawn daily log sheet per day

ELD Trip Planner

HOS-compliant routes and daily logs for property-carrying drivers

العربية

Trip details

Try an example

Current location

S Chicago, IL

Pickup location

P Dallas, TX

Dropoff location

D Los Angeles, CA

Current cycle used (hrs)

30

40 of 70 hours left in the 8-day cycle

Start time

10/02/2026 06:00 AM

Home terminal time. The driver starts fully rested.

Log sheet details (optional)

Plan trip

Property-carrying driver, 70 hrs / 8 days

Fuel at least every 1,000 miles

1 hour each for pickup and drop-off

TOTAL DISTANCE

2,404 mi

967 mi + 1,438 mi

DRIVING TIME

42h 15m

11h 30m door to door

ARRIVAL

Tue, Oct 6

9:30 PM, drop-off complete

LOG SHEETS

5 days

3 × 10-hr rest, 2 × fuel, 2 × 30-min break, 1 × 34-hr restart

CYCLE AFTER TRIP

6h 30m / 70h

Start

Pickup

Drop-off

Fuel stop

30-minute break

10-hour rest

34-hour restart

To pickup

967 mi · 17h 15m driving

I 90 → I 57 → I 55 → I 40 → I 440 → I 30

To drop-off

1,438 mi · 25h driving

I 30 → I 20 → I 10 → I 10 EXPR

The planned route for Chicago → Dallas → Los Angeles with 30 cycle hours used: 2,404 miles over 5 days.

Stack

Part	Technology
Backend	Django 5 + Django REST Framework, stateless
Frontend	React 19 + TypeScript + Vite
Map and routing	Leaflet, OpenStreetMap tiles, OSRM, Photon — all free, no API key
Log sheet	SVG drawn from the day's data

Hours-of-service rules applied

Rules follow the FMCSA Interstate Truck Driver's Guide to Hours of Service (49 CFR 395).

Rule	Limit	At the limit
Driving	11 hours after 10 hours off	10-hour rest (sleeper berth)
Driving window	14 hours from coming on duty	10-hour rest (sleeper berth)
Break	30 min after 8 hours of driving	30-minute break
Cycle	70 hours in 8 days	34-hour restart
Fuel	Every 1,000 miles at most	30-minute on-duty stop
Pickup, drop-off	1 hour each	On duty, not driving

How the schedule is built

The planner simulates the trip the way a driver runs it. It computes the time left before each limit, drives until the nearest one, adds the stop that limit requires, and repeats until the route is finished. The timeline is then cut at each midnight into one log per day.

Trip schedule

Stops and rests required by the hours-of-service rules

FRI, OCT 2

6:00 AM – 6:15 AM	● Pre-trip inspection	15m
	Chicago, IL · On duty (not driving)	
6:15 AM – 2:15 PM	● Drive 448 mi	8h
	Chicago, IL → Steele, MO · Driving	
2:15 PM – 2:45 PM	● 30-minute break	30m
	Steele, MO · Off duty	
2:45 PM – 5:45 PM	● Drive 168 mi	3h
	Steele, MO → Carlisle, AR · Driving	
5:45 PM – 3:45 AM	● 10-hour rest	10h
	Carlisle, AR · Sleeper berth	

SAT, OCT 3

3:45 AM – 4:00 AM	● Pre-trip inspection	15m
	Carlisle, AR · On duty (not driving)	
4:00 AM – 10:15 AM	● Drive 350 mi	6h 15m
	Carlisle, AR → Dallas, TX · Driving	
10:15 AM – 11:15 AM	● Pickup	1h
	Dallas, TX · On duty (not driving)	
11:15 AM – 11:45 AM	● Drive 29 mi	30m
	Dallas, TX → Fort Worth, TX · Driving	
11:45 AM – 12:15 PM	● Fuel stop	30m
	Fort Worth, TX · On duty (not driving)	
12:15 PM – 4:30 PM	● Drive 244 mi	4h 15m
	Fort Worth, TX → Big Spring, TX · Driving	
4:30 PM – 2:30 AM	● 10-hour rest	10h
	Big Spring, TX · Sleeper berth	

SUN, OCT 4

2:30 AM – 2:45 AM	● Pre-trip inspection	15m
	Big Spring, TX · On duty (not driving)	
2:45 AM – 10:45 AM	● Drive 460 mi	8h
	Big Spring, TX → Deming, NM · Driving	
10:45 AM – 11:15 AM	● 30-minute break	30m
	Deming, NM · Off duty	
11:15 AM – 2:15 PM	● Drive 173 mi	3h
	Deming, NM → Benson, AZ · Driving	
2:15 PM – 12:15 AM	● 10-hour rest	10h
	Benson, AZ · Sleeper berth	

MON, OCT 5

12:15 AM – 12:30 AM	● Pre-trip inspection	15m
	Benson, AZ · On duty (not driving)	
12:30 AM – 2:30 AM	● Drive 115 mi	2h
	Benson, AZ → Casa Grande, AZ · Driving	
2:30 AM – 3:00 AM	● Fuel stop	30m
	Casa Grande, AZ · On duty (not driving)	
3:00 AM – 5:00 AM	● Drive 115 mi	2h
	Casa Grande, AZ → Salome, AZ · Driving	
5:00 AM – 3:00 PM	● 34-hour restart	34h
	Salome, AZ · Off duty	

TUE, OCT 6

3:00 PM – 3:15 PM	● Pre-trip inspection	15m
	Salome, AZ · On duty (not driving)	
3:15 PM – 8:30 PM	● Drive 302 mi	5h 15m
	Salome, AZ → Los Angeles, CA · Driving	
8:30 PM – 9:30 PM	● Drop-off	1h
	Los Angeles, CA · On duty (not driving)	

The trip schedule: every stop with its time, place and duration.

Daily log sheets

Each sheet is drawn in SVG: the duty-status line on the 24-hour grid, the totals (always 24 hours), a bracket and city name for every stop in the Remarks, and the 70-hour recap. Sheets can be printed.

Drivers Daily Log

(24 hours)

10 / 03 / 2026
(month) (day) (year)

Original – File at home terminal.
Duplicate – Driver retains in his/her possession for 8 days.

From: Carlisle, AR

To: Big Spring, TX

623

Total Miles Driving Today

623

Total Mileage Today

Name of Carrier or Carriers

Main Office Address

Driver's Signature in Full

Truck/Tractor and Trailer Numbers or License Plate(s)/State (show each unit)

	Mid-night	1	2	3	4	5	6	7	8	9	10	11	Noon	1	2	3	4	5	6	7	8	9	10	11	Mid-night	Total Hours
1. Off Duty																										0
2. Sleeper Berth																										11.25
3. Driving																										11
4. On Duty (not driving)																										1.75
Remarks																										= 24
Shipping Documents:																										
DVL or Manifest No. or Shipper & Commodity																										

Recap:

Complete at end of day

70 Hour / 8 Day Drivers

Enter name of place you reported and where released from work and when and where each change of duty occurred. Use time standard of home terminal.

12.75

On duty hours today, total lines 3 & 4

54

A. Total hours on duty last 8 days including today

16

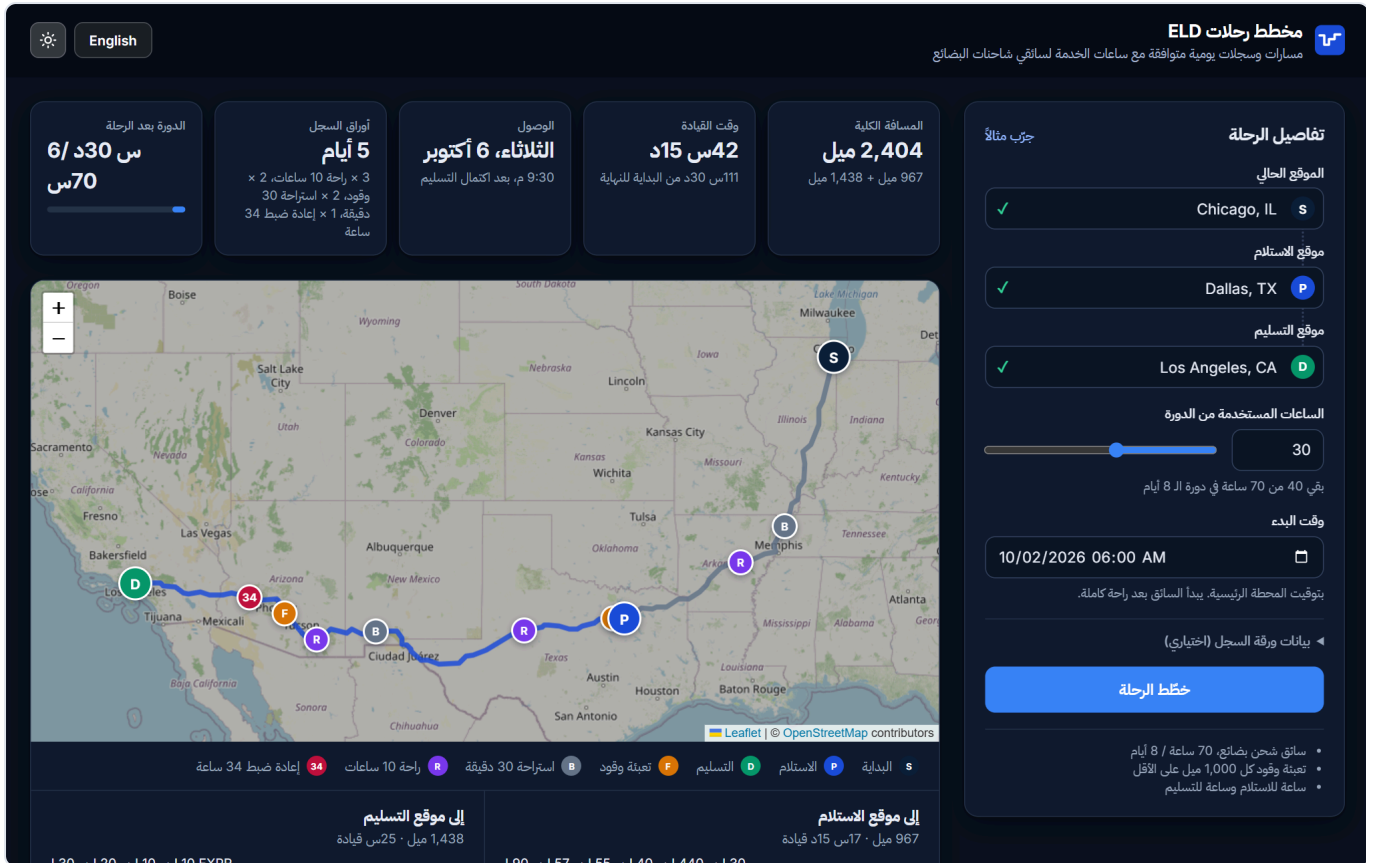
B. Total hours available tomorrow (70 hr. minus A)

* If you took 34 consecutive hours off duty you have 70 hours available.

Day 2 of the example trip.

Languages and themes

The interface is available in English and Arabic (right to left), with light and dark themes. The log sheet stays in English, as it reproduces a US DOT form.



Arabic interface in the dark theme.

Testing

Ten automated tests pass. Every generated plan is replayed by an independent checker that asserts each rule on each segment, across 60 combinations of distance, cycle hours and start time.

Assumptions and limitations

- Given: property-carrying driver, 70 hrs / 8 days, no adverse conditions, fuel every 1,000 miles, 1 hour each for pickup and drop-off.
- Added: the driver starts fully rested; a 15-minute pre-trip inspection starts each shift; a fuel stop takes 30 minutes; times are on 15-minute marks.
- "Current cycle used" is one number, so those hours are kept for the whole trip and recovered only by a 34-hour restart. This is conservative.
- The split sleeper-berth provision is not used.